

AI Time Table Generator Using Python, Streamlit and Genetic Algorithm

Submitted By

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Abstract

The AI Time Table Generator is an intelligent scheduling system developed using Python and Streamlit. The project automatically creates optimized and clash-free timetables using a Genetic Algorithm.

Manual timetable generation is difficult because teachers, rooms, classes, and subjects must all be arranged without conflicts. This project automates the process and reduces human effort.

The system uses Artificial Intelligence concepts, Data Structures, and optimization techniques to generate efficient schedules.

Introduction

Timetable scheduling is one of the most important tasks in educational institutions. Creating schedules manually often causes problems such as:

- Teacher clashes
- Room conflicts
- Unbalanced schedules
- Time wastage
- Human errors

This project solves these problems using Artificial Intelligence.

The system generates automatic timetables based on user inputs such as:

- Subjects
- Teachers
- Rooms
- Days
- Time slots

The project uses a Genetic Algorithm to continuously improve timetable quality.

Objectives

The objectives of this project are:

- Generate automatic timetables
 - Eliminate teacher clashes
 - Eliminate room conflicts
 - Optimize scheduling
 - Reduce manual work
 - Implement Artificial Intelligence techniques
 - Create an interactive Streamlit GUI
 - Store data permanently using SQLite
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Technologies Used

Technology	Purpose
Python	Main programming language
Streamlit	Frontend GUI
SQLite	Database storage
Pandas	Data display
Random Module	Random timetable generation
Genetic Algorithm	AI optimization

Data Structures Used

Lists

Used to store:

- Subjects
- Days
- Rooms
- Time slots

Dictionaries

Used to store:

- Teacher mapping
- Timetable data

Nested Dictionaries

Used to represent timetable structure. Example:

Day → Time Slot → Lecture Details

System Architecture

Frontend

The frontend is developed using Streamlit.

Users can:

- Enter subjects
- Enter teachers
- Enter rooms
- Generate timetable
- View generated timetable

Backend

The backend is implemented in Python.
It handles:

- Timetable generation
- Clash detection
- Fitness evaluation
- Genetic Algorithm operations

Database

SQLite database stores:

- Teachers
- Subjects
- Rooms
- Timetables

This allows permanent data storage.

Genetic Algorithm

The project uses a Genetic Algorithm because timetable scheduling is a Constraint Satisfaction Problem.

The algorithm works through multiple generations and continuously improves the timetable.

Working of Genetic Algorithm

Step 1 — Population Creation

Generate multiple random timetables.

Step 2 — Fitness Function

Evaluate timetable quality.

Step 3 — Selection

Choose best timetables.

Step 4 — Crossover

Combine good timetables.

Step 5 — Mutation

Apply small random changes.

Step 6 — Repeat

Repeat process until optimized timetable is achieved.

Fitness Function

The fitness function evaluates timetable quality.

Formula:

$$\text{Fitness} = 100 - (\text{Clashes} \times 10)$$

Higher fitness means:

- Better timetable
- Fewer conflicts
- More optimization

Constraints Used

The system checks the following constraints:

- Teacher clash detection
- Room clash detection
- Duplicate lecture checking
- Subject frequency management
- Lab room allocation

- Break timing management

Streamlit GUI

The project uses Streamlit for GUI development.

Features:

- User-friendly interface
 - Interactive forms
 - Timetable generation button
 - Fitness score display
 - Dynamic timetable table
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Project Workflow

Step 1

User enters timetable information.

Step 2

Frontend sends data to backend.

Step 3

Python backend generates random schedules.

Step 4

Genetic Algorithm optimizes timetable.

Step 5

Best timetable is selected.

Step 6

Final timetable is displayed in Streamlit.

Step 7

Data is saved into SQLite database.

Advantages of the System

- Saves time
 - Reduces manual effort
 - Eliminates clashes
 - Generates optimized schedules
 - Easy to use
 - Scalable system
 - Intelligent scheduling
-

Limitations

- Basic version handles limited constraints
 - Large institutions may require advanced optimization
 - Requires proper input formatting
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Future Enhancements

Future improvements may include:

- Teacher preference system
- Multiple department support
- Exam timetable generation
- Attendance integration
- PDF export
- Email notifications
- Cloud database support
- Login system

- Admin dashboard
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Results

The system successfully generates optimized and clash-free timetables.

The Genetic Algorithm improves timetable quality generation by generation.

The Streamlit GUI makes the system interactive and user-friendly.

Conclusion

The AI Time Table Generator demonstrates how Artificial Intelligence can solve real-world scheduling problems efficiently.

Using Python, Streamlit, SQLite, and Genetic Algorithms, the system automatically creates optimized timetables while minimizing conflicts.

The project combines:

- Artificial Intelligence
- Data Structures
- GUI development
- Database integration
- Optimization techniques

This makes the project practical, scalable, and suitable for educational institutions.

Final Project Statement

“This project uses Artificial Intelligence and Genetic Algorithm techniques to automatically generate optimized and clash-free academic timetables.”

Create Basic Timetable Structure

```
DAYS = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday"]

TIME_SLOTS = [
    "9-10",
    "10-11",
    "11-12",
    "1-2",
    "2-3"
]

SUBJECTS = [
    "Math",
    "Physics",
    "Programming",
    "English"
]

TEACHERS = {
    "Math": "Ali",
    "Physics": "Ahmed",
    "Programming": "Sara",
    "English": "Ayesha"
}

ROOMS = ["R1", "R2", "Lab1"]
```

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Create Random Timetable

```
import random

def create_random_timetable():
    timetable = {}

    for day in DAYS:
        timetable[day] = {}

        for slot in TIME_SLOTS:
            subject = random.choice(SUBJECTS)
            teacher = TEACHERS[subject]
            room = random.choice(ROOMS)

            timetable[day][slot] = {
                "subject": subject,
                "teacher": teacher,
                "room": room
            }

    return timetable
```

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Print Timetable

```
def print_timetable(timetable):
    for day, slots in timetable.items():
        print(f"\n{day}")

        for slot, details in slots.items():
            print(
                slot,
                details["subject"],
                details["teacher"],
                details["room"]
            )
```

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Clash Detection

```
def calculate_clashes(timetable):
    clashes = 0
    for slot in TIME_SLOTS:
        teachers_used = []
        rooms_used = []

        for day in DAYS:
            lecture = timetable[day][slot]

            teacher = lecture["teacher"]
            room = lecture["room"]

            if teacher in teachers_used:
                clashes += 1
            else:
                teachers_used.append(teacher)

            if room in rooms_used:
                clashes += 1
            else:
                rooms_used.append(room)

    return clashes
```

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Fitness Function

```
def fitness(timetable):  
    clashes = calculate_clashes(timetable)  
  
    return 100 - (clashes * 10)
```

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Generate Population

```
def create_population(size):  
    population = []  
  
    for i in range(size):  
        timetable = create_random_timetable()  
        population.append(timetable)  
  
    return population
```

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Selection

```
def selection(population):  
    population.sort(key=fitness, reverse=True)  
  
    return population[:2]
```

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Crossover

```
def crossover(parent1, parent2):  
    child = {}  
  
    for day in DAYS:  
        if random.random() > 0.5:  
            child[day] = parent1[day]  
        else:  
            child[day] = parent2[day]  
  
    return child
```

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Mutation

```
def mutate(timetable):
    day = random.choice(DAYS)
    slot = random.choice(TIME_SLOTS)

    new_subject = random.choice(SUBJECTS)

    timetable[day][slot]["subject"] = new_subject
    timetable[day][slot]["teacher"] = TEACHERS[new_subject]

    return timetable
```

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Genetic Algorithm Main Engine

```
def genetic_algorithm(generations=100, population_size=10):
    population = create_population(population_size)

    for generation in range(generations):
        parents = selection(population)

        new_population = []

        for _ in range(population_size):
            child = crossover(parents[0], parents[1])

            child = mutate(child)

            new_population.append(child)

        population = new_population

    best = selection(population)[0]

    return best
```

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Run Project

```
best_timetable = genetic_algorithm()

print_timetable(best_timetable)

print("Fitness:", fitness(best_timetable))
```

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...

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Monday
9-10 Physics Ahmed R2
10-11 English Ayesha R1
11-12 Physics Ahmed R1
1-2 Programming Sara R1
2-3 English Ayesha R2

Tuesday
9-10 Physics Ahmed Lab1
10-11 English Ayesha Lab1
11-12 Physics Ahmed R2
1-2 English Ayesha R2
2-3 Programming Sara R2

Wednesday
9-10 Physics Ahmed R1
10-11 Math Ali R2
11-12 Programming Sara R1
1-2 Math Ali Lab1
2-3 Math Ali R2

Thursday
9-10 Math Ali Lab1
10-11 Programming Sara R2
...
11-12 Programming Sara R2
1-2 Physics Ahmed R1
2-3 English Ayesha R1
Fitness: -80

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Frontend View

Generated Timetable

	Day	Time Slot	Subject	Teacher	Room
5	Tuesday	9-10	Physics	Ahmed	Lab2
6	Tuesday	10-11	Math	Ali	R2
7	Tuesday	11-12	English	Ayesha	Lab2
8	Tuesday	1-2	Physics	Ahmed	R2
9	Tuesday	2-3	Physics	Ahmed	Lab1
10	Wednesday	9-10	Math	Ali	R2
11	Wednesday	10-11	Physics	Ahmed	Lab2
12	Wednesday	11-12	English	Ayesha	R2
13	Wednesday	1-2	English	Ayesha	Lab2
14	Wednesday	2-3	Programming	Sara	R3
15	Thursday	9-10	Physics	Ahmed	R2